



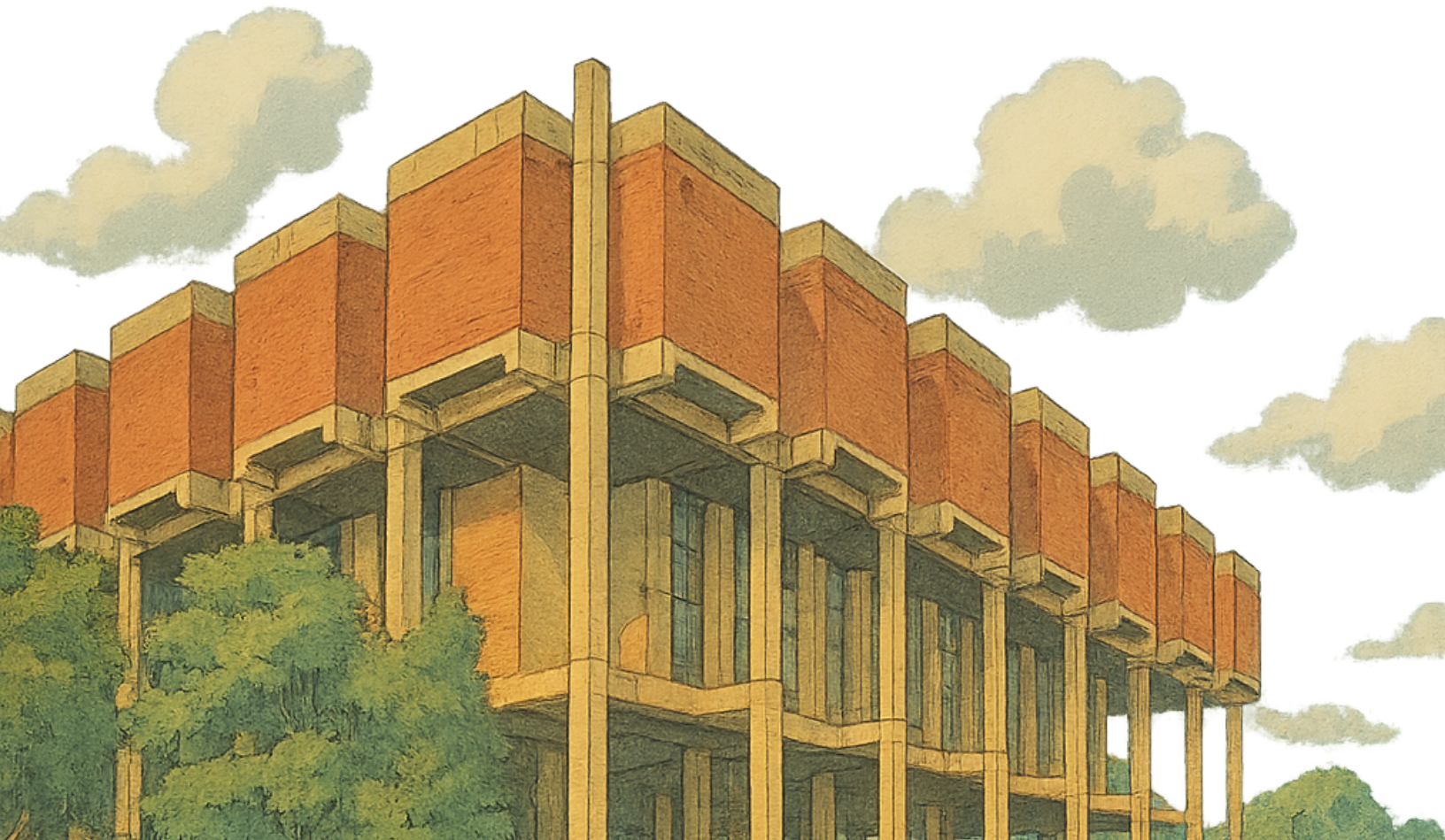
Indian Institute of Technology Kanpur

Department of Computer Science & Engineering

Placement Brochure

Session
2026-2027

The Recruiter's Guide
Students' Placement Office





Department of Computer Science & Engineering

Indian Institute of Technology, Kanpur

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About Us

The Computer Science & Engineering Department at IIT Kanpur is the first department in India to start Computer Science education. The department started in August 1963 with an IBM 1620 system--a novelty then, even in many North American and European universities.

Today, many of the nation's leading experts, educationists, and consultants in computer science are alumni of this department. Currently, the department has a faculty of 33 whose interests span almost all areas of computer science.

The department runs BTech, MTech, MS(R), and Ph.D. programs and is known for its academic rigor, excellence, and cutting-edge research with several sponsored projects and consultancies. Some students convert their B.Tech. program into a dual degree program, which results in students getting both a B.Tech and an M.Tech. degree at the end of 5 years. The department has also started the M.S. program, which is uniquely tailored for deep domain expertise, combining intensive thesis-driven research with high-impact software systems development. MS graduates bring specialised problem-solving skills, production-grade engineering experience, and advanced academic rigour to core industry roles."



Welcome Message



Prof. Surender Baswana

**Head of Department Computer Science
& Engineering
Indian Institute of Technology, Kanpur**

"I am proud to introduce the exceptional talent nurtured within our Computer Science and Engineering department. Our curriculum is meticulously designed to provide a strong foundation in computing while incorporating emerging and advanced topics. This ensures our students are always at the forefront of technological innovation.

Our graduates are not just engineers, they are visionaries and leaders, ready to make significant impacts in various industries. From core computing roles to positions in financial software, algorithmic trading, and E-commerce, our alumni are thriving and driving success in global companies. Additionally, many have ventured into entrepreneurship, founding and contributing to groundbreaking startups worldwide.

I invite you to visit our department and witness the unique, stimulating environment where your next top talent is being prepared for success. Our commitment to excellence guarantees that hiring from our department will bring unparalleled expertise and innovation to your organization."



Major Degree Programs

B.Tech

A rigorously structured undergraduate program that builds a formidable foundation in computer science. The curriculum is continuously aligned with modern industry requirements, producing engineers ready to tackle the most demanding technical challenges from day one.

BT-MT (Dual Degree)

The integrated five-year program culminates in the award of both B.Tech and M.Tech degrees. Students are introduced to core engineering disciplines during their undergraduate years, followed by a dedicated year of research contributing to their postgraduate specialization.

M.Tech

An intensive postgraduate program culminating in a research-focused final year. Students engage heavily in advanced group projects and master complex theoretical and applied concepts, making them ideal for high-level software and systems engineering roles.

Ph.D.

This prestigious degree is awarded to students committed to research careers. Unlike other programs, it emphasizes innovation and discovery over traditional coursework.



Major Degree Programs

M.S. (By Research)

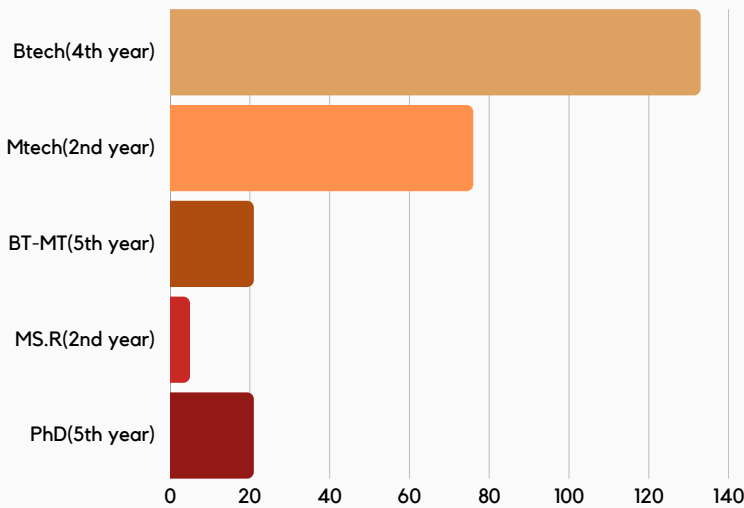
A highly specialized program tailored for deep domain expertise. Candidates undergo rigorous, thesis-driven research often tied to live, industry-sponsored projects in areas like AI, Systems, and Cybersecurity. By dedicating extended time to complex problem-solving rather than standard coursework, these graduates bring production-grade engineering skills and unparalleled technical depth to core industry roles.

eMasters in Cyber Security

An executive-level program designed to build advanced capabilities in digital infrastructure and threat mitigation. By bridging practical industry experience with cutting-edge cybersecurity research, these candidates are uniquely equipped to lead strategic security initiatives and safeguard enterprise systems at top organizations.



About Students



Number of final year students
(2025-2026)

Recent Achievements

- **Best Software Award (2026):** Amit Bhasita, Khushwant Kaswan, and Tippireddy Yashwanth (Class of 2026) won for developing **DAVi**—a secure, air-gapped data analytics platform.
- **Outstanding PhD Thesis Award (2026):** Dr. Madhavan Venkatesh for research in Algorithmic Arithmetic Geometry (Guided by Prof. M. Agrawal & Prof. N. Saxena).
- **Manas Mandal Best PhD Thesis Award (2026):** Dr. Utsav Singh for research on Hierarchical Strategies for Complex Control Tasks (Guided by Prof. S. Simon & Prof. V. Namboodiri).
- **Global Coding Excellence:** IIT Kanpur team "facelessmen3.0" represented the institute at the ICPC 2023 World Finals.

Distinguished Alumni

- **Prof. Rajeev Alur** (BT/CSE/1987) : Professor of Computer Science and the Founding Director of ASSET.
- **Mr. Ambarish Kenghe** (MT/CSE/1998) was appointed as the Group CEO of Angel One.
- **Prof. Pushpak Bhattacharyya** (MT/CSE/1986) : Chairman of RBI's committee to develop the Framework for Ethical AI (FREE-AI) in the financial sector.
- **Dr. Eshan Chattopadhyay** (BT/CSE/2011): Professor at Cornell University.
- **Dr. Prateek Jain** (BT/CSE/2004): Senior Research Staff at Google AI.
- **Prof. Ashish Goel** (BT/CSE/1994): Professor at Stanford University.
- **Mr. Rajnish Kumar** (BT/CSE): Co-Founder & Group Co-CEO of ixigo (India's leading AI travel platform) and Guest of Honour for Convocation 2026.
- list is long ...



Lab Facilities



Network

- The CSE lab has a 1 Gbps switched network using a tree topology. All servers and client systems have 1 Gbps NICs.
- A CISCO Catalyst 3750G Series Layer-3 switch serves as the backbone.
- Lab connects to the campus network via a router.
- Connectivity to the Institute Computer Center is through fiber optic cable.
- Network spans both CSE buildings using 50 switches (24 ports each, 1 Gbps).
- Two dedicated server rooms house a variety of servers, including project-specific systems.



Cloud

- CSE lab has deployed one cloud "**Vyomkesh**",
- It has currently 20 compute nodes having dual Xeon 6 core CPUs with 48 GB RAM each.
- It supports 240 physical (480 logical) cores with 960 GB RAM.



Hardware Lab

- The hardware lab in the CSE department is a state of the art lab equipped for embedded computing.
- The lab provides several FPGA-based stations for hardware programmability



Lab Facilities



Robotics Lab

- Vicon Vero Motion Capture System for indoor robot localization has been installed in this Lab



SCADA Cyber Security Test Bed

- A flexible SCADA test bed with Schneider industrial hardware and software is installed. Supports research in cybersecurity for critical infrastructure.
- Replicates power distribution automation (three-phase & single-phase).
- Includes real field devices:
 - Energy meters, protection relays, modular control system
 - GPS time sync unit (IRIG-B based), IEC61850 Ethernet switches
 - Unified Threat Management hardware, industrial-grade HMI
 - Full SCADA system for monitoring and control



Sun Grid

- Sun Grid has been established in the department with funding from Sun Microsystems, USA.
- The grid has 20 workstations based on Opteron (64-bit AMD processor) and it is available to all the users who want to do research and explore the area of High Performance Technical Computing.
- The department is recognized as the "Sun Regional Academic and Research Partner for Excellence in Grid Computing".



Lab Facilities



CDIS

- The Center for Developing Intelligent Systems - is an R&D Center within IIT Kanpur
- It is dedicated to rapid development and prototyping of intelligent software systems, geared towards solving problems arising within the Indian ecosystem.



C3i Center

- Cyber security is now a national security issue due to digital control in critical infrastructure.
- Systems like power grids, water systems, nuclear plants, railways, and air traffic rely on networked software control.
- These are vulnerable to cyber-attacks by nation-states, terrorists, or criminals.
- IIT Kanpur's Centre for Cyber Security (funded by SERB/DST) focuses on protecting critical infrastructure.
- The centre develops India's first cyber security test bed for such systems.
- Works closely with NCSC and government agencies to report threats and solutions.
- Trains students and utility professionals in cyber defence.
- Collaborates with global partners (USA, Israel) and organizes events to build awareness and research culture.



Faculty List and Expertise

Anshu Yadav

Theoretical Cryptography and Post Quantum Cryptography

Amey Karkare

Compilers, Functional Programming, Program Analysis & Code Optimization, Programming Languages and Education

Anil Seth

Logic, Theoretical Computer Science retrieval

Ashutosh Modi

Natural Language Processing, Machine Learning and Affective Computing

Debapriya Basu Roy

Hardware Security, VLSI for Cryptography and Post Quantum Cryptography

Adithya Vadapalli

Applied Cryptography, Private Information retrieval, Secure Multi party computation, Zero knowledge Proofs and Oblivious RAMs

Amitangshu Pal

Wireless and Sensor Networks, Sensing and Communication for Internet of Things (IoTs) and Building IoT Solutions for Smart Cities

Angshuman Karmakar

Post Quantum Cryptography, SideChannel attacks and Computation on encrypted data and Cryptology

Arnab Bhattacharya

Databases, Data Mining, Information Retrieval, Natural Language Processing and Artificial Intelligence

Faculty List and Expertise

Debadatta Mishra

Operating Systems, Virtualization and
Cloud Computing and Computer Networks

Hamim Zafar

Computational Biology, Machine
Learning and Bioinformatics

Mainak Chaudhuri

Computer Hardware and Architecture

Nisheeth Srivastava

Human Factors in Computing,
Computational Cognitive Science and
Computational Social Science

Piyush Rai

Machine Learning and Bayesian Statistics

Gunjan Kumar

Sublinear Algorithms, Approximation
Algorithms, Statistical Learning Theory
and Information Theory

Indranil Saha

Application of formal methods to
embedded and cyberphysical systems
and Robotics

Manindra Agrawal

Computational Complexity Theory,
Security, Theoretical Computer Science
and Algorithms

Nitin Saxena

Computational Complexity Theory,
Algebra, Algebraic Geometry,
Theoretical Computer Science and
Algorithms

Faculty List and Expertise

Preeti Malakar

High Performance Computing, Scalable
Parallel Communication and Workflow
Optimization

Roop Aparajita Subhra Purushottam

Optimization, Statistical Learning
Theory and Machine Learning

Rajat Mittal

Computational Complexity and Quantum
Computing and Semidefinite
Programming

Pritam Choudhury

Programming Languages, Type Theory,
Static Analyses, Language-Based
Security, Formal Verification, Logic, and
Formal Methods in Indian Knowledge
Traditions

Satyadev Nandakumar

Theoretical Computer Science,
Algorithmic Information Theory and
Computable Analysis

Priyanka Bagade

internet of things, computer vision, deep
learning, and cyber-physical systems

Raghunath Tewari

Computational Complexity Theory, Graph
Theory, Theoretical Computer Science
and Computational Geometry

Rajat Moona

Computer Architecture, VLSI Design,
Operating Systems, Embedded Systems,
Security, Smart Cards and RFID

Sanjeev Saxena

Parallel Processing, VLSI, Data Structures,
Algorithms, Heuristics, Graph Theory and
Computational Geometry

Faculty List and Expertise

Sayak Ray Chowdhury

Reinforcement Learning, Multi-armed bandits, Differential Privacy and Language Model Alignment

Sruti S Ragavan

Human-Computer Interaction, Software Engineering, End-user Programming, Humanistic Aspects of Computation

Sumit Ganguly

Databases, Data Streaming and Algorithms

Surender Baswana

Graph Algorithms, Dynamic Algorithms and Randomized Algorithms and Algorithms

Swarnendu Biswas

Programming Languages, Compilers, Program Analysis, Software systems and High Performance Computing

Soumya Dutta

High Performance Computing and Visualization, Visual Computing and Computer Vision, Big Data Visual Analytics

Subhajit Roy

Formal Methods, Artificial Intelligence, Software Engineering, Programming Languages and Compiler Optimization

Sunil Simon

Game Theory, Distributed Systems and Temporal Logics and Verification

Sutanu Gayen

Foundations of Machine Learning and Probabilistic Algorithms

Faculty List and Expertise

Urbi Chatterjee

Hardware Security, Physically Unclonable Functions, Secure Authentication Protocols Design and Internet of Things Security

Prateek Jain

Visiting Faculty

Principal Scientist / Director at Google DeepMind India.

Amartya Sanyal

Adjunct Faculty

Assistant Professor in Machine Learning in the Department of Computer Science in the University of Copenhagen.

Shashank Srivastava

Adjunct Faculty

Assistant Professor of Computer Science at UNC Chapel Hill.

Gaurav Sharma

Visiting Faculty

Computer Vision, Face Image Analysis, Visual Analytics, Machine Learning and Deep Learning

Rajesh Kumar Gupta

Visiting Faculty

Professor in the Department of Computer Science and Engineering at the University of California, San Diego.

Pravesh K. Kothari

Adjunct Faculty

Assistant Professor in Computer Science Department, Princeton..

Snigdha Chaturvedi

Adjunct Faculty

Assistant Professor at UC-Santa Cruz.



Industrial Engagements

Research/Project Topic	Organisation
Establish an academic partnership program that aims to foster innovation in areas such as earth and cloud computing as well as data-driven content creation	
Collaborate on the advancement of digital forensic technology for medical devices	
Establish a state-of-the-art Advanced Innovative Integrated Laboratory at the ITITI Doon Sanskrit School premises.	 Information Technology Institute for the Tribes of India
Jointly develop sector-specific Cyber Security Capability Maturity Models for the National Critical Information Infrastructure Protection Center (NCIIPC)	 IHUB NTIHAC Foundation
Development and generation of a range of semantically equivalent queries and prompts for general QnA systems	
Methods for Identifying and Redacting Protected Health Information in Clinical Notes	
Establish the Intel India Research Fellowship Agreement with the aim of funding for a candidate to pursue a research project.	
Use of e-Governance technologies such as blockchain to safeguard data integrity in records management, integrated financial management systems, e-coupons, electronic health records, and the Janadhar residents' data repository	



Industrial Engagements

Research/Project Topic	Organisation	Professor name
Video analytics for smart surveillance systems		Amitangshu Pal
seamless transition to post-quantum cryptography		Angshuman Karmakar
Interior Design		Nisheeth Srivastava
Establishing the UP Digital Health Stack		Nisheeth Srivastava
DAVi: An Intelligent Data Analytics and Visualization Framework		Soumya Dutta & Purushottam Kar
Digital Public Goods for AI in Health		Nisheeth Srivastava
Reviewing status of telemedicine hubs in Uttar Pradesh		Nisheeth Srivastava
Samar Abhilekh: Controlled Digital Lending for MoD History Division records		Nisheeth Srivastava
HT cable fault prediction.		Sayak Ray Chowdhury
Deep Learning-Based Time Series Forecasting of Short-Term Gas Demand		Sayak Ray Chowdhury

List is long ..



Recent Research & Developments

Cognitive Fatigue in Autoregressive Transformers (ICML 2026)

- Formalized long-horizon generation degradation in LLMs as 'cognitive fatigue' and introduced the Fatigue Index (FI) as a model-agnostic diagnostic for runtime reliability monitoring.
- Analyzed 9 models (1B–13B parameters) to accurately predict task degradation (AUROC = 0.95) and repetition, revealing critical non-monotonic scaling behaviors in instruction-tuned models.

Canonicalizing Multimodal Contrastive Representation Learning (ICLR 2026)

- Proved that independently trained multimodal contrastive models (e.g., CLIP, SigLIP) share geometric relationships that can be aligned by a single orthogonal map across modalities.
- This theoretical breakthrough enables backward-compatible model upgrades without costly re-embedding, securing a highly competitive Oral Presentation at ICLR 2026.

LDAL: Dynamic Loss Function for Long-Tailed Learning

- Developed a novel Learning-Dynamics Aware Loss (LDAL) function to tackle extreme class imbalances in real-world computer vision datasets (e.g., medical imaging, fault detection).
- Adjusts class weights dynamically using Shannon entropy and inter-epoch regularization, setting new State-of-the-Art (SOTA) benchmarks on CIFAR-LT and ImageNet-LT.

DRAVYA: Guardrailed LLM Agent for WDN Analysis

- Developed a fully-promptable, low-code/no-code decision-support agent for end-to-end Water Distribution Network analysis under the Jal Jeevan Mission (Govt. of India).
- Implemented a 5-stage domain-aware guardrail pipeline with an auto-repair loop (≤ 3 re-prompts) to eliminate silent simulation failures, reducing user interaction steps by $>93\%$.



Recent Research & Developments

BODH.AI: Secure Benchmarking Platform for Healthcare ML

- Architected a self-hosted, Kaggle-like evaluation system enabling secure benchmarking of machine learning models on sensitive medical data.
- Implemented an isolated MLOps pipeline using FastAPI, React, Redis, and network-disabled read-only Docker containers to prevent data leakage and ensure zero-trust execution.

ISLKG: First Knowledge Graph for Indian Sign Language

- Building the first comprehensive Knowledge Graph for Indian Sign Language, curating over 19,000 dictionary videos covering 12,000+ words to supply missing linguistic structure.
- Developed a neuro-symbolic framework combining Spatio-Temporal Graph Convolutional Networks (ST-GCN) for keypoint pose tracking with knowledge graph embeddings to infer unseen signs.

DAVi: Secure Collaborative Data Analytics Platform

- Conferred the Best Software Award (2026) at the 59th IITK Convocation for developing a highly robust data visualization and analysis platform.
- Designed specifically for organizations dealing with sensitive data, the DAVi platform enables secure, air-gapped installations and effectively prevents vendor lock-in scenarios.

Blockchain-Based Urban E-Governance

- The CSE department, in collaboration with C3iHub, is spearheading the development of a secure Blockchain-Based TDR Portal for the Kanpur Development Authority (KDA).
- The platform utilizes blockchain technology to ensure the tamper-proof generation, transfer, and utilization of Transferable Development Rights, setting a new benchmark for state e-governance.



Pioneering Breakthroughs & National Impact

The AKS Primality Test (Polynomial-Time Primality)

- Solved one of the greatest open problems in computational number theory by proving deterministically that primality can be tested in polynomial time.
- Developed by Prof. Manindra Agrawal, Neeraj Kayal, and Prof. Nitin Saxena, this breakthrough won the Gödel Prize and the Fulkerson Prize, reshaping global algorithmic complexity.

Interdisciplinary Center For Cyber Security And Cyber Defence

- DST has sanctioned ₹14.43 crores for the Interdisciplinary Center for Cyber Security and Cyber Defense of Critical Infrastructure, led by Prof. Sandeep Shukla and Prof. Manindra Agrawal.
- This is the first center set up in India dedicated exclusively to pursuing and advancing the cyber security of national critical infrastructure.

JEE (JoSAA) Seat Allocation Framework

- Designed the core algorithmic framework for joint seat allocation across all IITs, NITs, and CFTIs in India (led by Prof. Surender Baswana).
- This highly robust algorithm handles complex choices for hundreds of thousands of candidates annually, ensuring transparent, efficient, and error-free nationwide admissions.

TRINETRA: Unified Geospatial Perception System

- Developed an LLM-agent-routed meta-ensemble architecture combining Tri-YOLO, SAM, and GeoQwen for multi-modal satellite imagery interpretation (captioning, grounding, and VQA).
- Achieved state-of-the-art benchmark performance on VRSBench and advanced temporal change detection, winning the Silver medal at the Inter IIT Tech Meet 14.0 and accepted at IGARSS 2026.



Foundational Courses

- Mathematics for Computer Science I, II, III
- Computer Organisation, Computer Architecture
- Software Development and Operations
- Principles of Database Systems
- Operating Systems
- Compiler Design
- Theory of Computation
- Principles of Programming Languages,
- Computer Networks
- Computer Systems Security
- Fundamentals of Computing I, II
- Data Structures and Algorithms
- Algorithms II
- Sensing, Comm. and Networking for Smart Wireless Devices
- Introduction to Machine Learning
- Data Mining
- Indexing and Searching Techniques in Databases

Advanced Courses

- Large Language Models (LLMs)
- Blockchain Technology
- Statistical Natural Language Processing
- Deep Reinforcement Learning
- Probabilistic Machine Learning
- Generative Artificial Intelligence
- Advanced Topics in Algorithms and Data Structures
- Computer Vision and Image Processing
- Concurrent Data Structures and Algorithms
- Cryptographic Techniques for Privacy
- Preservation Quantum Computing,
- Topics in Distributed Systems,
- Parallel Computing Program Analysis
- Linux Kernel Programming
- Multi-core and Multiprocessor Architecture
- Programming for Performance
- Functional Programming
- IoT System Design
- Secure Memory Systems

*A detailed list of courses can be found [here](#)



Prominent Recruiters



Contact Us



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Faculty Placement Coordinator

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